

1. A random sample of size $n_1 = 16$ is selected from a normal population with a mean of 75 and a standard deviation of 8. A second random sample of size $n_2 = 9$ is taken from another normal population with mean 70 and standard deviation 12. Let X_1 and X_2 be the two sample means. Find:
 - (a) The probability that $X_1 - X_2$ exceeds 4
 - (b) The probability that $3.5 \leq X_1 - X_2 \leq 5.5$
2. PVC pipe is manufactured with a mean diameter of 1.01 inch and a standard deviation of 0.003 inch. Find the probability that a random sample of $n = 9$ sections of pipe will have a sample mean diameter greater than 1.009 inch and less than 1.012 inch.
3. From the data on the pH of rain in Ingham County, Michigan: 5.47, 5.37, 5.38, 4.63, 5.37, 3.74, 3.71, 4.96, 4.64, 5.11, 5.65, 5.39, 4.16, 5.62, 4.57, 4.64, 5.48, 4.57, 4.57, 4.51, 4.86, 4.56, 4.61, 4.32, 3.98, 5.70, 4.15, 3.98, 5.65, 3.10, 5.04, 4.62, 4.51, 4.34, 4.16, 4.64, 5.12, 3.71, 4.64 What proportion of the samples has pH below 5.0?
4. The following are 25 readings of proton lux in the 47-68 keV range: 2310, 2320, 2010, 10800, 2190, 3360, 5640, 2540, 3360, 11800, 2010, 3430, 10600, 7370, 2160, 3200, 2020, 2850, 3500, 10200, 8550, 9500, 2260, 7730, 2250.
 - (a) Find a point estimate of the mean proton flux in this time period.
 - (b) Find a point estimate of the standard deviation of the proton flux in this time period.
 - (c) Find an estimate of the standard error of the estimate in part (a).
5. Consider the probability density function $f(x) = c(1 + \theta x)$, $-1 \leq x \leq 1$.
 - (a) Find the value of the constant c .
 - (b) What is the moment estimator for θ ?
 - (c) Show that $\hat{\theta} = 3\bar{X}$ is an unbiased estimator for θ .
 - (d) Find the maximum likelihood estimator for θ .